EXPERIMENT 8

Contiguous Memory Allocation: BEST FIT

AIM:

To implement the Best Fit memory allocation algorithm and allocate memory blocks to processes efficiently.

PROCEDURE:

1. Start the Fedora operating system using Oracle VirtualBox.
2. Open the terminal and create a C program.
3. Enter the number of memory blocks and processes.
4. Read the sizes of memory blocks and processes.
5. For each process, find the smallest available memory block that can accommodate it.
6. Allocate the process to the selected block and update the remaining block size.
7. Repeat until all processes have been considered.
8. Display the allocation details and verify the output.

PROGRAM:

```
#include <stdio.h>
int main()
{
    int b[20], p[20], allocation[20];
    int nb, np, i, j, best;
    printf("Enter number of memory blocks: ");
    scanf("%d", &nb);
    printf("Enter memory block sizes:\n");
    for(i = 0; i < nb; i++)
        scanf("%d", &b[i]);
    printf("Enter number of processes: ");
    scanf("%d", &np);
    printf("Enter process sizes:\n");
    for(i = 0; i < np; i++)
        scanf("%d", &p[i]);

    for(i = 0; i < np; i++)
        allocation[i] = -1;

    for(i = 0; i < np; i++)
    {
        best = -1;

        for(j = 0; j < nb; j++)
        {
            if(b[j] >= p[i])
```

```
            {
                if(best == -1 || b[j] < b[best])
                    best = j;
            }
        }

        if(best != -1)
        {
            allocation[i] = best;
            b[best] -= p[i];
        }
    }

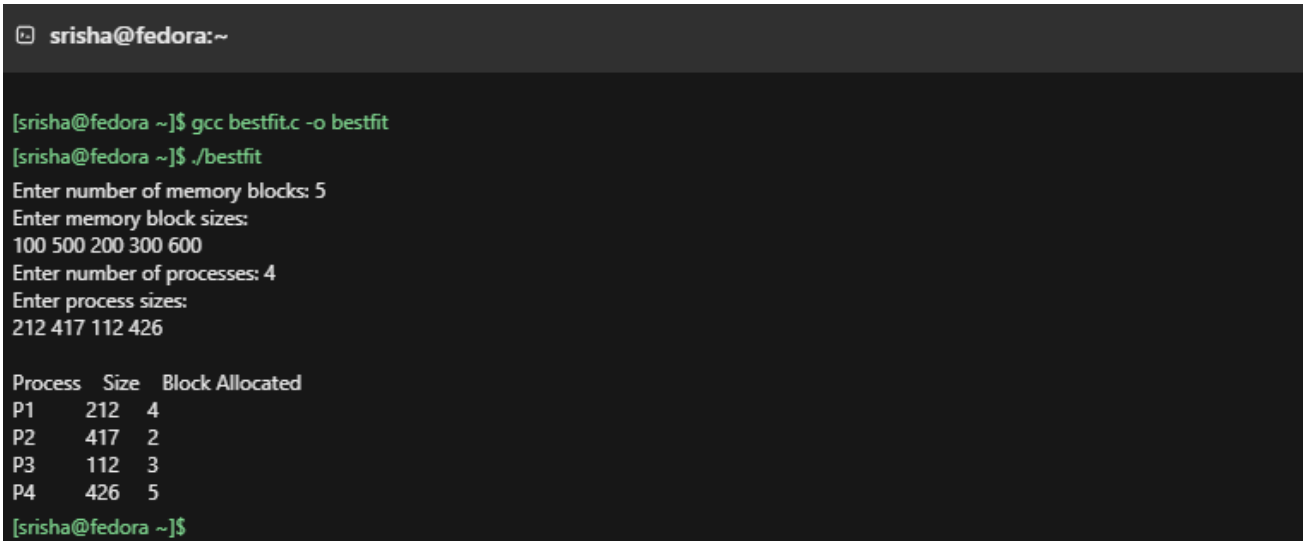
    printf("\nProcess\tSize\tBlock Allocated\n");

    for(i = 0; i < np; i++)
    {
        printf("P%d\t%d\t", i+1, p[i]);

        if(allocation[i] != -1)
            printf("%d\n", allocation[i]+1);
        else
            printf("Not Allocated\n");
    }

    return 0;
}
```

OUTPUT:



RESULT:

Thus, the Best Fit memory allocation algorithm was implemented and executed successfully in Fedora Linux, and memory blocks were allocated to the processes according to their requirements.